

DEVALIVE

Development Roadmap Document (DRD)

Version: 1.0

Document Status: Approved

Purpose:

This document defines the complete development roadmap for DevAlive.

It outlines:

- Development Phases
- Milestones
- Feature Priorities
- MVP Scope
- Post-MVP Enhancements
- Future SaaS Evolution

The objective is to provide a structured execution plan from initial development through future platform growth.

1. Roadmap Overview

DevAlive development is divided into:

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Phase 1
Project Foundation

Phase 2
Authentication System

Phase 3
Project Management System

Phase 4
Monitoring Engine

Phase 5
Analytics System
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Phase 6
Notification System

Phase 7
Frontend Dashboard

Phase 8
Marketing Website

Phase 9
Testing & QA

Phase 10
Deployment & Launch

2. MVP Definition

The Minimum Viable Product (MVP) includes:

User Authentication
Project Registration
Automated Monitoring
Monitoring Logs
Analytics Dashboard
Email Alerts
Settings Management
Landing Page

The MVP goal is:

Allow developers to monitor deployed projects and receive downtime alerts.

3. Phase 1 — Project Foundation

Objective:

Establish project structure and development environment.

Backend Tasks

- Initialize Node.js Project
- Configure Express
- Setup MongoDB Atlas
- Setup Environment Variables
- Create Folder Structure
- Configure Middleware

Frontend Tasks

- Initialize React + Vite
- Configure Tailwind CSS
- Setup Routing
- Create Folder Structure
- Configure Axios

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- Working Project Skeleton

4. Phase 2 — Authentication System

Objective:

Implement secure user authentication.

Backend

User Model

Register API

Login API

JWT Authentication

Password Hashing

Protected Middleware

Frontend

Register Page

Login Page

Authentication Context

Protected Routes

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Secure Authentication Flow

5. Phase 3 — Project Management System

Objective:

Allow users to manage monitored projects.

Backend

Create Project API

Update Project API

Delete Project API

Get Projects API

Frontend

Projects Page

Add Project Modal

Edit Project Modal

Delete Confirmation

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Complete Project Management Module

6. Phase 4 — Monitoring Engine

Objective:

Build the core DevAlive monitoring system.

Backend

Node Cron Scheduler

Health Check Service

Response Time Tracking

Failure Detection

Monitoring Logs

Statistics Updates

Test Cases

Healthy Endpoint

Offline Endpoint

Timeout

HTTP Errors

Deliverable

Fully Functional Monitoring Engine

7. Phase 5 — Analytics System

Objective:

Provide uptime and performance insights.

Backend

Dashboard Statistics APIs

Analytics APIs

Uptime Calculations

Response Time Metrics

Frontend

Analytics Page

Charts

Statistics Widgets

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Analytics Dashboard

8. Phase 6 — Notification System

Objective:

Notify users about project issues.

Backend

Notification Service

Email Service

Downtime Alerts

Recovery Alerts

Frontend

Notifications Page

Notification Center

Alert Indicators

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Automated Alert System

9. Phase 7 — Dashboard Experience

Objective:

Build the complete SaaS dashboard.

Screens

Dashboard

Projects

Project Details

Analytics

Logs

Notifications

Settings

Features

Responsive Design

Search

Filters

Pagination

Empty States

Loading States

Deliverable

Production Ready Dashboard

10. Phase 8 — Marketing Website

Objective:

Create a professional SaaS marketing experience.

Pages

Landing Page

Features

How It Works

Documentation

Sections

Hero

Problem

Solution

Features

Workflow

Dashboard Preview

Testimonials

CTA

Deliverable

Complete SaaS Website

11. Phase 9 — Testing & QA

Objective:

Verify platform quality and reliability.

Testing Areas

Authentication

Projects

Monitoring Engine

Notifications

Analytics

Responsive UI

Security

Tasks

Bug Fixes

Performance Optimization

Security Validation

Production Readiness Review

Deliverable

Release Candidate Build

12. Phase 10 — Deployment & Launch

Objective:

Deploy DevAlive to production.

Frontend

Deploy To Vercel

Backend

Deploy To Render

Database

MongoDB Atlas

Tasks

Configure Environment Variables

Verify Monitoring

Verify Notifications

Production Testing

Deliverable

Live Production Platform

13. Post-MVP Roadmap

Version 2.0

Features

Slack Notifications

Discord Notifications

Telegram Notifications

Webhook Alerts

Goal

Improve notification flexibility.

14. Version 3.0 Roadmap

Features

Team Workspaces

Organization Accounts

Shared Dashboards

Role Management

Goal

Support team collaboration.

15. Version 4.0 Roadmap

Features

Subscription Plans

Usage Limits

Billing System

Payment Integration

Goal

Commercial SaaS Expansion.

16. Version 5.0 Roadmap

Features

Redis Queues

BullMQ Workers

Distributed Monitoring

Multi-Region Monitoring

Goal

Enterprise-grade scalability.

17. Version 6.0 Roadmap

Features

AI Monitoring Insights

Failure Prediction

Performance Recommendations

Intelligent Alert Prioritization

Goal

AI-powered reliability platform.

18. Milestone Summary

Milestone 1

Authentication Complete

Milestone 2

Project Management Complete

Milestone 3

Monitoring Engine Complete

Milestone 4

Analytics Complete

Milestone 5

Notifications Complete

Milestone 6

Dashboard Complete

Milestone 7

Marketing Website Complete

Milestone 8

Testing Complete

Milestone 9

Production Launch

19. Success Criteria

The DevAlive MVP is considered successful when:

Users Can Register

Users Can Add Projects

Monitoring Runs Automatically

Downtime Is Detected

Alerts Are Sent

Analytics Are Visible

Dashboard Is Responsive

Platform Is Stable

20. Development Roadmap Summary

The DevAlive roadmap follows a phased development strategy that prioritizes:

- Core Monitoring Functionality
- User Experience
- Reliability
- Scalability
- Future SaaS Growth

The MVP focuses on solving the primary problem of keeping developers informed about the health and availability of their deployed applications, while future versions expand toward a complete developer reliability platform.